IBM FileNet P8 V5.5.8

Day 10

Deployment, Migration & Certification Preparation

FDM Deployment

Performance Tuning

C1000-074 Cert Prep

FDM | Performance Tuning | IBM C1000-074 Exam Prep

Rajesh Pasham | Final Day of 10

Learning Objectives – Day 10

FDM
Explain FDM's role in promoting configurations between Dev/Test/Prod environments

FDM
Export object store configs, workflows, classes, and security settings using FDM

FDM
Import FDM packages and validate all migrated components in ACCE

Perf
Apply CE performance tuning: JVM heap sizing, DB connection pools, content caching

Perf
Diagnose workflow bottlenecks and configure WAS trace logging for CE debugging

Cert
Recall FileNet P8 architecture, CE admin, workflow, security concepts for C1000-074

Cert
Answer 20 timed practice questions across all 5 C1000-074 exam domains

Lab
Complete full FDM export+import cycle and performance tuning on IBM FileNet V5.5.8

01

FileNet Deployment Manager (FDM)

Exporting · Importing · Environment Promotion · Migrated Components

FileNet Deployment Manager – Overview

What FDM DOES Migrate

- Document classes & property templates
- Choice lists & entry templates
- Workflow definitions (.pep files)
- Storage policies
- Event subscriptions & Code Modules
- Security templates & Marking Sets
- Empty folder hierarchy

What FDM does NOT Migrate

- Document binary content (files)
- LDAP user/group accounts
- WAS server configuration
- Object store physical paths
- JDBC data source settings
- Object-level ACLs on existing docs
- ICN desktop configurations

Environment Promotion Strategy

DEVELOPMENT

Developers create/modify classes, workflows, entry templates. FDM export after each sprint.

`C:\FDM\Dev\`

FDM Export

FDM Import

TEST / UAT

QA validates migrated config. Integration testing. UAT sign-off required before Prod.

`C:\FDM\Test\`

FDM Export

FDM Import

PRODUCTION

Controlled import in maintenance window. Rollback plan ready. No direct changes to Prod.

`C:\FDM\Prod\`

Key Rules: Never edit Production directly · Tag packages with version+date · Always create a rollback package BEFORE each import · Store packages in version control

FDM Import Modes & Post-Import Checklist

Import Modes

Create Only

Creates new objects only. Fails if object already exists. Use for first-time imports to empty OS.

Update Only

Updates existing objects. Fails if object not found. Use for applying changes to already-imported configs.

Create or Update

Creates OR updates — handles both. Most flexible. Recommended for repeated promotions.

Replace

Overwrites target object completely. Destroys target customizations. Use with extreme caution.

Post-Import Validation Checklist

- [ ] Document Classes present in ACCE > Data Design
- [ ] Property Templates present with correct data types
- [ ] Storage Policies present and assigned to classes
- [ ] Event Subscriptions present with Enabled status
- [ ] Workflow Definitions in Process Config Console
- [ ] Empty folder hierarchy recreated in Browse
- [ ] Re-assign LDAP groups (GUIDs differ per environment)
- [ ] Verify storage area physical paths exist on target
- [ ] Test an application operation end-to-end

02

Performance & Troubleshooting

CE Performance Tuning · Workflow Performance · Logging & Monitoring

Content Engine Performance Tuning

JVM Heap (WAS)

Initial: 512 MB (-Xms)
Max: 2048 MB (-Xmx)
GC Policy: -Xgcpolicy:gencon
Nursery: -Xmn256m
Path: WAS > server1 > JVM

DB Connection Pool

Min: 10 connections
Max: 50 connections
Timeout: 30 seconds
Path: WAS > JDBC > Data Sources > Connection Pool

Content Cache

Size: 2-4x daily retrievals
Location: Local SSD (not network)
Monitor: ACCE > Cache Statistics
Target hit ratio: > 80%
Separate cache per cluster node

CE SQL Optimization

Use PropertyFilter (never null)
Set maxItems limit always
Index custom search properties
Avoid leading % in LIKE queries
Use INFOLDER to scope searches

Logging, Monitoring & Common Issues

Key Log Files

SystemOut.log	CE startup, auth, JDBC, API errors — primary diagnostic log
SystemErr.log	Java exceptions and stack traces
CE Trace Log	CE-specific trace (enable in ACCE > Admin > Logging). Very verbose.
PE Server Log	Workflow routing and step execution events
FDM Log	FDM export/import operation results
DB2 diag.log	DB2 errors, JDBC timeouts, lock waits

```
Trace: *=INFO:com.filenet.api.*=FINE  →  Reset to *=INFO
after debugging
```

Common Issues & Resolutions

Slow retrieval — Fix: Increase cache, move to SSD

Cause: Cache miss / slow disk

CE API timeout — Fix: Increase Max Connections to 50+

Cause: DB pool exhausted

Workflow not launching — Fix: Check PE service, verify CE Connection Point

Cause: PE stopped / CP offline

FDM GUID conflict — Fix: Use Create or Update import mode

Cause: Object name conflict

LDAP auth fail — Fix: Re-configure VMM LDAP in WAS

Cause: Wrong Bind DN credentials

OutOfMemoryError — Fix: Increase -Xmx to 2048m or 4096m

Cause: JVM heap too small

03

Certification Preparation

IBM C1000-074 · 5 Exam Domains · 63 Questions · 75% Pass Score

IBM C1000-074 – Exam Domain Breakdown

63 Questions · 90 Minutes · 75% Pass Score (47 correct) · IBM Certified Administrator - FileNet Content Manager V5.5

20%	26%	20%	18%	16%

CPE, PE, ICN, WAS, DB2, LDAP/VMM, ACCE, FDM. Component roles and interactions. 3 required databases.

Object store creation. Document classes. Property templates. Storage areas. Storage policies. ACCE operations.

PE architecture. Connection Points. Isolated Regions. Queues. Work items. Event-driven workflow launch.

ACLs, ACEs, Deny precedence. Special identities. LTPA tokens. VMM LDAP. Inheritance. Marking Sets.

FDM packages. Import modes. What FDM migrates. Post-import steps. Environment promotion strategy.

C1000-074 – Must-Know Facts

3 DBs required	GCDDB (domain config), OSDB (object store metadata), PEDB (Process Engine data)
WAS port	9080 (HTTP) | 9443 (HTTPS) | 9060/9043 (Admin Console)
LTPA timeout	Default 30 minutes. WAS Admin Console > Security > Global Security > LTPA.
Connection Point	P8Domain > Process Engine > Connection Points. Required for CE-PE integration.
FDM migrates	Classes, properties, workflows, policies, subscriptions, security templates.
	Write-Once-Read-Many (WORM). Required

PE port	32768 (RMI/IIOP) for CE-to-PE communication
ACE Deny rule	Deny ALWAYS overrides Allow. No exceptions. Order doesn't matter.
#CREATOR-OWNER	Resolves to the user who created each specific document. Per-object identity.
Isolated Region	Groups workflows for performance isolation on a single PE server. Default = Region 1.
FDM does NOT	Binary content, LDAP accounts, WAS config, physical storage paths.
	http://host:9080/fnsmis/browser/ ... JSON

Day 10 Lab Exercises

LAB-10-01 · 45 min
FDM Export

- Access FDM in ACCE > Tools > Deployment Manager
- Create full export package for TrainingOS
- Configure scope: classes, workflows, policies, subscriptions
- Review export log and inspect manifest.xml
- Verify ZIP package at C:\Day10Labs\fdm_packages\

LAB-10-02 · 40 min
FDM Import & Validate

- Create pre-import rollback package first
- Import package with Create or Update mode
- Review import log for warnings/errors
- Validate all components in ACCE post-import
- Identify and resolve any post-import issues

LAB-10-03 · 40 min
Performance Tuning

- Review and update WAS JVM heap (512/2048 MB)
- Tune DB2 connection pool (min:10, max:50)
- Restart WAS and verify CE comes back up
- Enable CE API trace and review log output
- Monitor cache statistics in ACCE

LAB-10-04 · 55 min
Certification Drill

- Domain-weighted exam summary review
- 20-question timed practice (40 min)
- Score and review answer explanations
- Identify weak domains for further study
- Complete course evaluation

IBM FileNet P8 V5.5.8

10-Day Training Complete

Congratulations, Rajesh Pasham!

Days 1-2	Days 3-4	Days 5-6	Days 7-8	Days 9-10
ECM Foundation, Architecture	CE Fundamentals, Metadata Modeling	Storage, Security, Process Engine	Workflow Dev, IBM Content Navigator	API Integration, FDM, Certification Prep

Ready for IBM C1000-074 Certification Exam

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